

7. Integration

What students need to learn:

Integration of e^x , $\frac{1}{x}$, $\sin x$, $\cos x$.

To include integration of standard functions such as $\sin 3x$, $\sec^2 2x$, $\tan x$, e^{5x} , $\frac{1}{2x}$.

Students should recognise integrals of the form

$$\int \frac{f'(x)}{f(x)} dx = \ln f(x) + c.$$

Students are expected to be able to use trigonometric identities to integrate, for example, $\sin^2 x$, $\tan^2 x$, $\cos^2 3x$.

Evaluation of volume of revolution.

$\pi \int y^2 dx$ is required, but *not* $\pi \int x^2 dy$.

Students should be able to find a volume of revolution, given parametric equations.

Simple cases of integration by substitution and integration by parts. These methods as the reverse processes of the chain and product rules respectively.

Except in the simplest of cases the substitution will be given.

The integral $\int \ln x dx$ is required.

More than one application of integration by parts may be required, for example $\int x^2 e^x dx$.

Simple cases of integration using partial fractions.

Integration of rational expressions such as those arising from partial fractions, e.g. $\frac{2}{3x+5}$, $\frac{3}{(x-1)^2}$.

Note that the integration of other rational expressions, such as $\frac{x}{x^2+5}$ and $\frac{2}{(2x-1)^4}$ is also required (see above paragraphs).

Analytical solution of simple first order differential equations with separable variables.

General and particular solutions will be required.
